



The House Always Wins: Machine Learning and Anomaly Detection in U.S. Congressional Stock Transactions

Sebastián Burgermeister

Carlos Huerta Araujo

Iván Robles Urquiza

Master in Economics — Universidad de San Andrés (UdeSA)

Big Data

January 2026

Abstract

We propose a machine learning framework to detect the use of private information in financial markets, based on comparing out-of-sample predictive accuracy between models using only public data and models augmented with variables capturing the behavior of potentially informed agents. We apply this approach to the trading activity of U.S. Congress members using a panel of stocks over 2012–2024. Adding congressional trading features improves abnormal return predictions, but only when using nonlinear models; linear specifications show no gain. The improvement is modest, statistically significant, and robust to placebo tests. A complementary unsupervised analysis identifies individual trades with anomalous characteristics. These trades earn higher abnormal returns, consistent with the hypothesis that some legislators trade on information not yet reflected in prices.

Contents

1	Introduction	3
2	Institutional Framework and Related Literature	5
3	Data	6
3.1	Data Sources	6
3.2	Variable Construction	8
3.3	Descriptive Analysis	9
4	Predictive Content of Congressional Trades	11
4.1	Model Specification and Evaluation Strategy	11
4.2	Statistical Significance	13
4.3	What the Models Capture	15
5	Anomaly Detection	18
5.1	Detection of Anomalous Trades	20
5.2	Validation: Abnormal Returns of Anomalous Trades	23
5.3	Characterization of Anomalous Trades	26
6	Conclusion	30

1 Introduction

Stock returns have long been recognized as difficult to predict, despite decades of effort and breakthroughs in the finance literature. This well-documented fact is often interpreted as evidence of market efficiency: if prices reflect all available information, future returns cannot be predicted with current data. However, the inability to predict returns admits a different interpretation: in settings where one suspects the presence—and usage—of private information, it should not be possible to predict returns using only public data. In that sense, one could ask whether variables that proxy for private information can improve prediction, and what this would suggest about the economic mechanisms that drive stock returns.

Grossman and Stiglitz (1980) formalize a fundamental tension within this interpretation. If prices fully reflected all available information, no agent would have an incentive to bear the cost of acquiring it, since informed trading could not be compensated. But if no one acquires information, prices cannot be informative in the first place. The resolution is that markets must be informationally inefficient to a degree that sustains equilibrium: informed agents earn just enough to justify their costs, and prices incorporate information only gradually and imperfectly. This framework generates a sharp empirical prediction—agents who acquire information at low cost should be able to exploit it most effectively, precisely because their required compensation is small—. Members of the United States Congress represent a natural case—and the one we examine in this work—: their access to market-relevant information is not purchased through research effort but obtained as a byproduct of their institutional role, through committee oversight, policy deliberations, and interactions with regulated industries. If any group of agents can profitably trade on private information in equilibrium, legislators are among the most plausible candidates.

Testing this hypothesis requires a framework that can distinguish genuine informational content from noise. Mullainathan and Spiess (2017) draw a sharp distinction between prediction and inference in applied econometrics: machine learning excels at the former but cannot establish causation. However, some economic theories—market efficiency, information asymmetry—make claims that are fundamentally about predictability. A model that fails to predict better when augmented with variables capturing private information provides evidence against the hypothesis that such information is being used; a model that does predict better provides evidence in favor. In this sense, prediction is not merely a technical exercise but a direct test of economic theory.

The empirical benchmark for such a test is well established. Welch and Goyal (2008) show that a wide array of variables proposed in the literature fail to predict equity returns out of sample, with many producing negative R-squared statistics. Gu, Kelly, and Xiu (2020) apply various machine learning methods to a comprehensive set of predictors,

obtaining modest but positive out-of-sample performance; in their findings, nonlinear models outperform linear specifications. A model that predicts worse than the historical mean would have the newly included variables adding noise rather than information. In this way, the benchmark of unpredictability is not arbitrary: it reflects the hypothesis that returns behave like a random walk.

We use this logic to test for the usage of private information. We construct a test for the information content of a particular set of signals by comparing predictive accuracy across nested models. The baseline specification includes only publicly available market variables, while the augmented specification adds variables designed to capture a theoretically motivated source of private information. If the augmented model improves out-of-sample prediction, this provides evidence that the additional variables contain information not reflected in prices. Even if small by conventional standards, the magnitude of improvement may be economically meaningful: Campbell and Thompson (2008) show that modest gains in return predictability can translate into substantial improvements in portfolio performance.

The empirical setting we examine is the trading activity of members of the United States Congress. Beyond the theoretical motivation outlined above, this setting offers two practical advantages. The STOCK Act of 2012 mandates periodic disclosure of trades, generating a public record that can be matched to stock-level outcomes. Additionally, the institutional context provides guidance on which variables might proxy for private information—such as committee assignments, seniority, and timing of disclosure—enabling theory-driven feature construction rather than pure data mining. Prior work has documented abnormal returns on congressional portfolios using event-study methods, though the interpretation of these findings remains debated (Ziobrowski, Boyd, et al. 2011; Ziobrowski, Cheng, et al. 2004).

Translating these institutional features into predictive variables, however, raises a methodological challenge. Machine learning algorithms are designed to optimize predictive accuracy, not to recover interpretable parameters; and the space of potential predictors is vast: raw variables can be transformed, interacted, lagged, or aggregated in countless ways. Selecting among all possible constructions risks overfitting to noise. Regularization disciplines this process by penalizing complexity, but it does not resolve the question of which variables to include in the first place. Fortunately, economic theory provides guidance. In a regularized framework, including a variable that lacks true predictive power carries limited cost: penalization shrinks its coefficient toward zero, and its contribution to out-of-sample error is minimal. Including a variable that does carry information, however, allows the model to detect and exploit that signal. This asymmetry favors building an extensive set of theoretically motivated predictors, even when the relevance of any particular variable is uncertain *ex ante*.

We apply this principle to construct features capturing the direction, magnitude, timing, and political characteristics of congressional trades at the stock-month level. Variables measure not only whether politicians are buying or selling, but also who is trading (committee assignments, seniority, party affiliation) and how they are trading (size of positions, delay between transaction and disclosure, whether multiple legislators act in coordination). Each variable reflects a mechanism suggested by prior literature on information asymmetry in political markets. The data construction process is itself a form of prior, embedding economic reasoning into the structure of the prediction problem before any algorithm is applied.

We find that models incorporating congressional trading variables produce higher out-of-sample R-squared than models using market information alone. The improvement is modest in absolute terms but statistically significant according to tests designed for nested forecast comparison. Most notably, the gains appear only in nonlinear specifications—tree ensembles and gradient boosting—while linear models show no improvement or even degradation. This suggests that the relationship between congressional trading and future returns involves interactions or thresholds that linear methods cannot capture. A placebo test in which randomly generated variables replace the congressional features confirms that the improvement is not merely a result of increased model complexity.

2 Institutional Framework and Related Literature

Prior to 2012, members of Congress were not required to publicly disclose their stock transactions in a timely manner, with reporting delays ranging from 5 to 17 months as reported by Karadas (2019). The *STOCK Act* of 2012 changed this by prohibiting legislators from trading on material nonpublic information and requiring electronic disclosure of their transactions within 30 to 45 days. The present study relies on the institutional framework established by the *STOCK Act*, which requires members of Congress to disclose their stock transactions electronically. These mandatory filings, available from 2012 onward, constitute our primary data source, though they come with two notable limitations: transaction amounts are reported in ranges rather than exact values, and there is a delay between the trade date and its publication, meaning that the information reaches the market with a lag.

The possibility that U.S. legislators exploit informational advantages derived from their office to earn superior stock returns has been documented in the literature. Ziobrowski, Boyd, et al. (2011) and Ziobrowski, Cheng, et al. (2004) constructed portfolios replicating the trades disclosed by senators and representatives, finding monthly abnormal returns of approximately 85 and 55 basis points, respectively. These results suggest that legislators may have incorporated material nonpublic information obtained through their government roles into their investment decisions. However, while these studies document superior

performance, they do not directly test whether congressional trades contain private information —outperformance could reflect skill, luck, or access to better financial advice rather than informational advantages—. Our approach differs: rather than measuring portfolio returns, we test whether congressional trading activity has predictive content for future stock returns beyond what is captured by public market information.

Recent evidence suggests that the STOCK Act had a deterrent effect. Karadas, Schlosky, and Hall (2021) show that politicians’ aggregated stock trades predicted market returns prior to the legislation, yet this predictive ability disappeared after the introduction of the act, suggesting that the informational advantage was institutional rather than the result of individual stock-picking ability. Despite this improvement, aggregate evidence indicates that congressional trading has resumed outperforming the broader market. Portfolios constructed by replicating the disclosed trades of members of Congress have consistently beaten the S&P 500 in recent years.¹

Moreover, several controversial episodes have cast further doubt on the effectiveness of the current regulatory framework. The most salient occurred in 2020, when several senators were implicated in a trading scandal ahead of the COVID-19 market crash.² These types of episodes, which intensify during periods of high economic uncertainty as documented by Mazzarisi et al. (2024), motivate the need to develop analytical tools capable of detecting potentially anomalous trading patterns.

Our work also connects with a broader political economy literature that examines transparency and accountability as mechanisms for disciplining elected officials. Ferraz and Finan (2008) study the effect of random audits on municipal corruption in Brazil and find that public disclosure of audit results significantly reduces the reelection probability of mayors involved in irregularities, particularly in municipalities with greater local media presence. For the U.S. context, Snyder Jr and Strömberg (2010) show that press coverage of House representatives has direct effects on accountability: voters in districts with less media coverage are less informed about their representatives, and those representatives in turn exert less effort on behalf of their constituents.

3 Data

3.1 Data Sources

This study constructs a novel, integrated dataset that links congressional stock transactions to detailed market and political context. Rather than relying on a single source, we merge three complementary data streams: (i) transaction-level disclosures made by members of

¹“Members of Congress Outperformed S&P 500,” *Yahoo Finance*, December 15, 2024, [link](#).

²“El FBI investiga a un influyente senador por uso de información privilegiada,” *Ámbito Financiero*, March 2020, [link](#).

the United States Congress, (ii) asset-level financial market data capturing returns, risk exposures, and trading conditions, and (iii) biographical and institutional characteristics of legislators. This integration allows us to situate each trade within both its financial environment and the political position of the trader, enabling a richer analysis of informational content than would be possible using any source in isolation.

Transaction-level data on stock trades by members of the U.S. Congress come from Quiver Quantitative, a platform that aggregates disclosures required under the *Stop Trading on Congressional Knowledge* (STOCK Act) of 2012. This legislation requires members of Congress to publicly report their financial transactions. Quiver Quantitative collects and systematizes these records from the U.S. Office of Government Ethics, the Senate Office of Public Records, and the Office of the Clerk of the House of Representatives.

For each transaction, the data include: asset ticker, transaction date, type of operation (purchase, sale, or exchange), reported amount range, and public disclosure date. A limitation of the regulatory regime is that amounts are reported in intervals (e.g., \$1,001–\$15,000 or \$15,001–\$50,000) rather than exact values, as the law does not require disclosure of precise dollar amounts or number of shares. Additionally, regulations allow a lag of 30 to 45 days between the transaction execution date and its public disclosure.

Financial market data are obtained from Yahoo Finance. For each ticker present in congressional transactions, daily prices were downloaded, along with fundamental firm information (market capitalization, price-to-book ratio, EV/EBITDA). From these data, we construct a set of variables characterizing asset behavior prior to each transaction: returns at various frequencies, momentum measures across multiple horizons, realized volatility, liquidity indicators, and abnormal volume. Additionally, we estimate risk factor exposures using the CAPM model and the Fama-French three-factor model, employing 252-day rolling windows. As outcome variables, we calculate cumulative abnormal returns (CAR) over 30, 60, and 90-day windows following each transaction, using three alternative specifications: market-adjusted return (S&P 500), CAPM model, and Fama-French three-factor model.

Biographical and political information on members of Congress comes from two complementary sources. First, the congress-legislators GitHub repository,³ which integrates data from ProPublica and GovTrack. Second, additional information from Ballotpedia on estimated net worth and demographic characteristics.

Variables obtained include: full name, date of birth, gender, party affiliation, state of representation, type of office (representative or senator), years of seniority, estimated net worth, and religion. Additionally, we collected committee assignments for each legislative period from information available on Wikipedia.

³<https://github.com/unitedstates/congress-legislators>

The dataset covers the period from June 2012 to January 2026, spanning from the implementation of the STOCK Act to the present. The original database includes transactions across multiple asset types (stocks, options, bonds, ETFs, cryptocurrencies); for this study, only common stock purchase and sale transactions are retained. After this filter, the sample contains 80,080 transactions corresponding to 274 unique members of Congress across 4,683 tickers.

This transaction-legislator level database constitutes the input for the anomaly detection analysis. For the return prediction analysis, transactions were collapsed to the ticker-month level, yielding a panel of approximately 32,000 observations where each row represents an asset-month with at least one congressional trade. In this aggregation process, count variables are summed (number of purchases, sales, transactions by members of informative committees), continuous legislator variables are averaged (seniority, net worth, political power index), and ratios are constructed to normalize each dimension by the total transactions for that ticker-month.

3.2 Variable Construction

Following the recommendations of Mullainathan and Spiess (2017), predictor variables were explicitly constructed rather than delegating their discovery to the algorithm. The authors argue that economic theory and domain knowledge play a crucial role in guiding where the algorithm searches for structure: explicitly creating variables suspected to be relevant reduces the complexity cost associated with regularization, thereby improving predictive performance.

From the original variables in the databases, aggregated indicators were constructed at the ticker-month level to capture collective signals of congressional behavior. Key variables include: a sentiment index measuring the difference between purchases and sales normalized by total transactions; the proportion of transactions made by members of informationally sensitive committees; and the proportion of transactions executed by committee *chairs* or *ranking members*, who presumably have greater access to privileged information. An indicator was also constructed identifying ticker-months where the net position is positive, at least one trader belongs to an informative committee, and at least one holds a leadership position on that committee.

Additionally, variables capturing coordination patterns among legislators were constructed, such as the number of transactions where two or more members of Congress trade the same asset on the same day, and an indicator of potentially suspicious coordination when this occurs among members of the same informative committee. Timing measures are also included, such as the average lag between transaction date and public disclosure, as well as variables characterizing the market context in which legislators trade—for

example, whether they trade against the asset's recent *momentum*.⁴

3.3 Descriptive Analysis

This section provides an overview of the main characteristics of the sample, examining the temporal distribution of transactions, their composition by legislative chamber and party affiliation, the most frequently traded assets, and the distribution of trade amounts.

Figure 1 presents the temporal evolution of congressional transactions between 2012 and 2026. Trading activity exhibits sustained growth, with a notable increase starting in 2017 and peaks in 2020 and 2025. The 2020 spike coincides with the volatility generated by the COVID-19 pandemic, a period during which several members of Congress faced public scrutiny for trades executed following confidential briefings on the health crisis. Throughout the entire period, purchases slightly exceed sales.

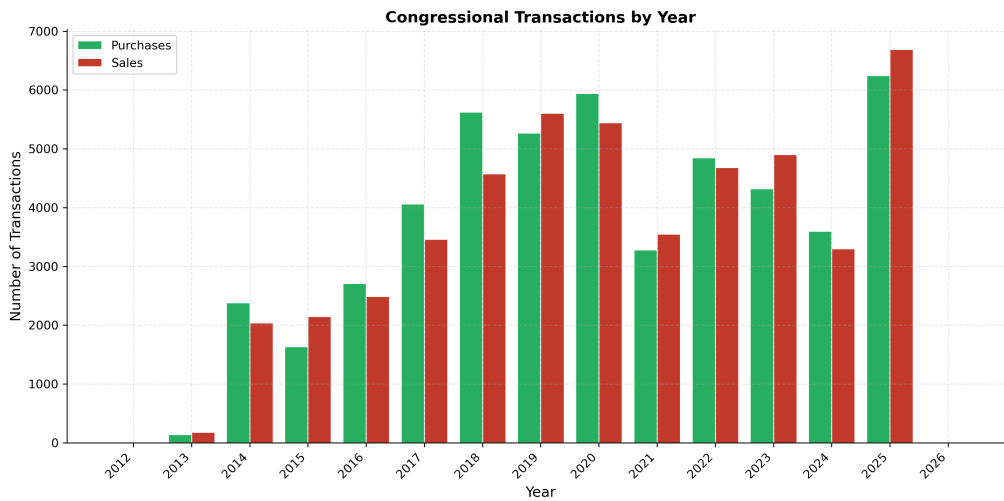


Figure 1: Congressional transactions by year

Turning to the composition of trades, Figure 2 displays the distribution by legislative chamber and party affiliation. Panel (a) shows that the House of Representatives accounts for 90% of transactions, while the Senate represents only 10%. This difference is partially explained by the larger number of representatives (435) compared to senators (100), although the proportion of trades exceeds the proportion of members, suggesting higher activity among representatives. Panel (b) reveals that Democrats and Republicans exhibit identical levels of activity (approximately 49,700 transactions each), while independents represent a marginal fraction of the sample. The partisan balance suggests that trading behavior is not concentrated in one party, a relevant consideration for the analysis that follows.

⁴In total, 70 congressional trading variables were constructed.

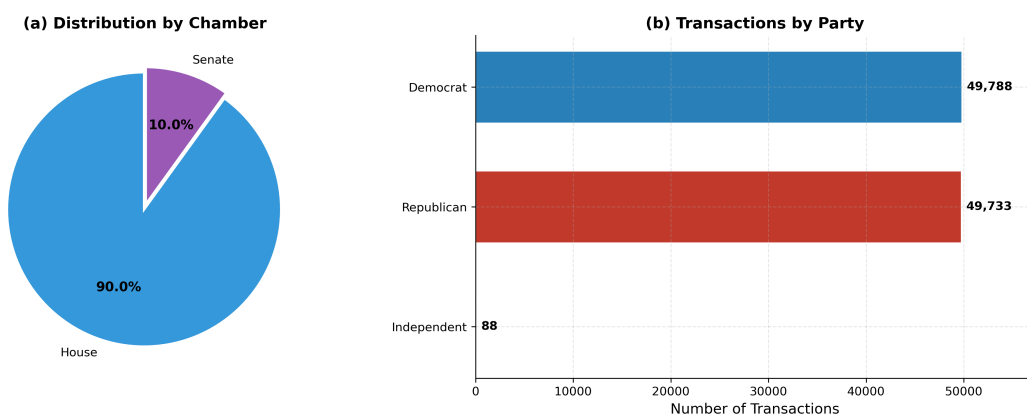


Figure 2: Distribution of trades by legislative chamber and party affiliation

Regarding the assets traded, Figure 3 illustrates the ten most frequently traded tickers. Large technology companies dominate the ranking: Microsoft (MSFT) and Apple (AAPL), followed by Amazon (AMZN), Nvidia (NVDA), and Alphabet (GOOGL). This concentration in high-capitalization, highly liquid firms is consistent with diversified portfolios, though it also coincides with sectors frequently subject to congressional regulation.

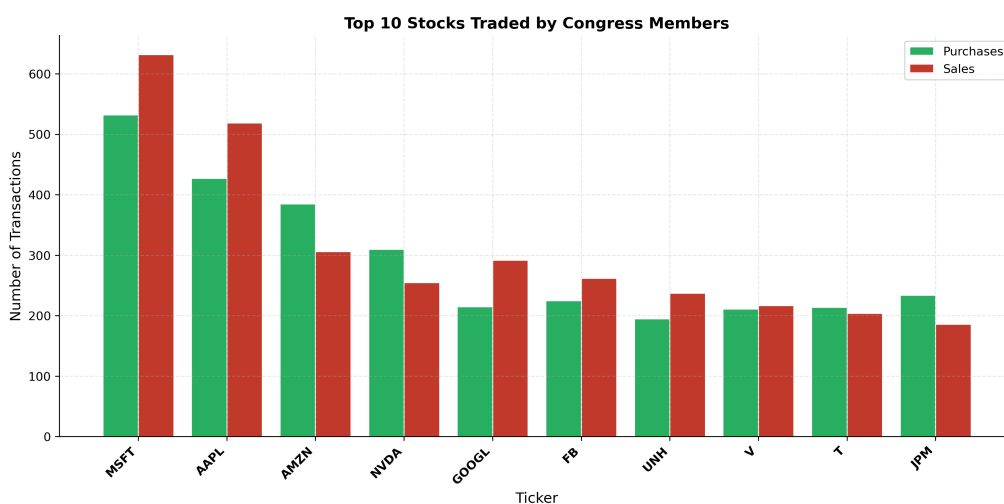


Figure 3: Top 10 stocks traded by members of Congress

Finally, Figure 4 shows the distribution of transactions by amount range. Approximately 73% of trades fall within the lowest bracket (\$1,001–\$15,000), and 89% are concentrated in amounts below \$50,000. High-value transactions (exceeding \$100,000) represent less than 5% of the total. This distribution reflects both the reporting constraints of the STOCK Act, which requires disclosure in ranges rather than exact amounts, and potential trade-splitting strategies that may obscure the true magnitude of positions.

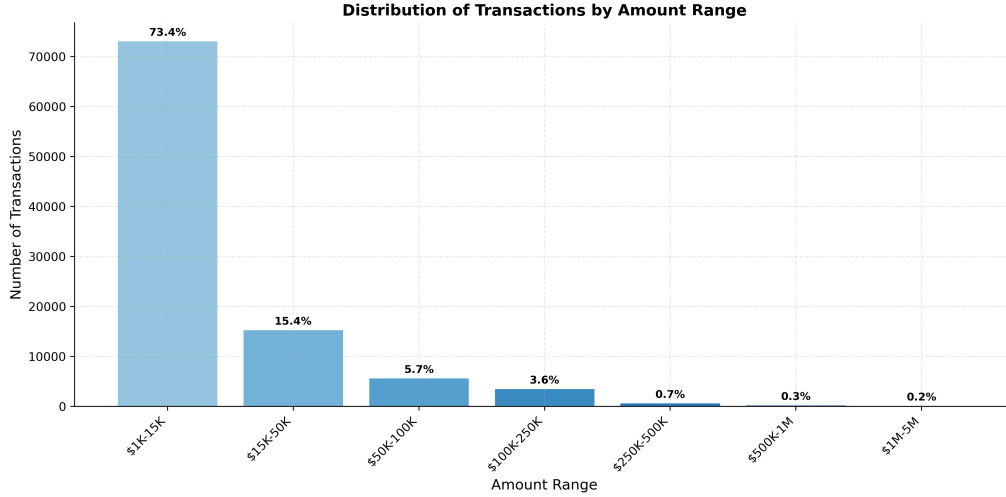


Figure 4: Distribution of transactions by amount range

4 Predictive Content of Congressional Trades

4.1 Model Specification and Evaluation Strategy

We estimate each model two times: once using only public market variables, and once adding the congressional trading features described in Section 3. Both specifications are then evaluated out of sample through an expanding window that begins with 36 months of training data and adds one month at a time, yielding 113 test periods between 2015 and 2024. Finally, we compare whether the augmented specification has more predictive power than the base specification.

We consider six distinct models spanning a range of functional form assumptions: Ordinary least squares provides an unrestricted linear benchmark. Ridge regression adds ℓ_2 penalization, shrinking coefficients toward zero without excluding variables, which helps when predictors are correlated but does not perform variable selection. Lasso imposes ℓ_1 penalization, driving some coefficients exactly to zero and thus selecting a subset of predictors. Elastic net combines both penalties, offering a middle ground that can better handle correlated groups of variables. These four methods all assume that the relationship between predictors and returns is linear and additive. Random forest relaxes this assumption, as it constructs an ensemble of shallow decision trees trained on a bootstrap sample with a random subset of features, and averages their predictions. Averaging reduces variance while the tree structure allows for interactions and threshold effects without requiring the researcher to specify them. Gradient boosting takes a different approach to ensemble construction: rather than building trees independently, it fits them sequentially, with each tree targeting the residual errors of the previous ones. A low learning rate controls the contribution of each step, and early stopping prevents the model from fitting noise. Both tree-based methods can capture nonlinear relationships between

congressional trading patterns and subsequent returns, which happens to be relevant to our results.

Table 1 reports the out-of-sample performance for OLS and the two tree-based models across both specifications. Results for Ridge, Lasso, and Elastic Net are omitted for brevity but follow patterns consistent with OLS: regularized linear models either show no improvement or slight degradation when congressional features are added. The metric is R^2_{OOS} , defined as one minus the ratio of squared prediction errors to the variance of realized returns. A value of zero corresponds to predicting the historical mean, that is to say, the naive forecast implied by a random walk. Positive values indicate the model captures some structure in returns; negative values indicate it does worse than guessing the mean.

Table 1: Out-of-Sample Predictive Performance

Model	Specification	R^2_{OOS} (%)	RMSE	Correlation	Hit Rate (%)
OLS	Base	0.34	0.0928	0.077	–
	Augmented	–0.18	0.0931	0.067	–
Random Forest	Base	1.41	0.0923	0.122	54.46
	Augmented	1.41	0.0923	0.123	54.54
Gradient Boosting	Base	0.84	0.0926	0.103	53.83
	Augmented	0.85	0.0926	0.104	53.84

The linear models either fail to benefit from the congressional features or are hurt by their inclusion. OLS and Ridge both show a decline in R^2_{OOS} when the augmented features are added. With 70 additional variables and a linear specification, these models overfit to patterns in the training set that do not generalize. Lasso and Elastic Net tell a related but distinct story: they produce identical results across both specifications, meaning the regularization path sets all congressional coefficients to zero. In the Lasso’s view, no single congressional variable carries enough linear signal to justify its inclusion over the penalty threshold. However, the exclusion of congressional variables in this model is itself informative: it does not mean the variables are irrelevant; but that any relevance they have does not take a linear form.

The tree-based models reinforce the idea of congressional variables having a non-linear form. Random Forest achieves an R^2_{OOS} of 1.41% in both specifications, with a marginal improvement when congressional features are included⁵. Gradient Boosting moves from 0.84% to 0.85%. The improvements are small in absolute terms (as they are

⁵The rounded figures in Table 1 mask a small improvement visible at higher decimal precision. The Clark-West test below confirms that this difference is statistically significant for Gradient Boosting, as the test is designed to detect improvements in the structure of forecast errors even when summary metrics appear identical.

fractions of a percentage point) but the direction is consistent, and the levels themselves are worth contextualizing. In the return prediction literature, out-of-sample R^2 values near 1% are considered meaningful. Gu, Kelly, and Xiu (2020) report similar magnitudes for individual stock return prediction using tree-based methods and a much larger set of predictors, and they document that such levels of predictability translate into economically significant portfolio gains. Campbell and Thompson (2008) make the general argument that even modest improvements in R^2_{OOS} can generate substantial utility gains for mean-variance investors. Our base models, using only standard market characteristics, already reach these levels—a finding consistent with the broader literature on cross-sectional return predictability—.

4.2 Statistical Significance

To assess whether the improvement from congressional features is statistically distinguishable from zero, we use the Clark and West (2007) test. This test is designed for nested model comparison under the null hypothesis that the smaller model is correctly specified. Under the null, the augmented model includes irrelevant parameters that add noise to its predictions, making raw MSE comparisons biased against it. The Clark-West adjustment corrects for this bias by subtracting the squared difference between the two forecasts from the loss differential.

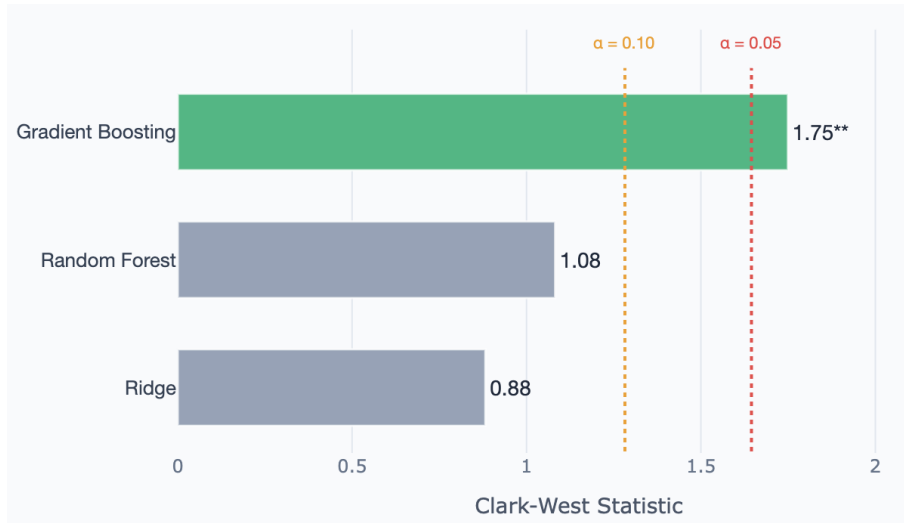


Figure 5: Clark-West test statistics for the augmented versus base specification. The null hypothesis is that the base model (market variables only) is sufficient. Dashed lines indicate critical values at the 10% and 5% significance levels.

Figure 5 reports the results for three models. Gradient Boosting produces a Clark-West statistic of 1.75, which exceeds the 5% critical value of 1.645. Random Forest reaches 1.08 and Ridge 0.88, both below conventional thresholds. The test thus provides formal evidence that congressional trading features improve prediction under Gradient Boosting,

while the evidence for other models is suggestive but not conclusive at standard confidence levels.

This pattern where only one model class reaches statistical significance warrants discussion. The Clark-West test evaluates the marginal contribution of the new variables within a given model. If the model cannot represent the functional form through which those variables affect returns, the test will fail to reject even when the variables do carry information. The fact that Gradient Boosting, which builds its predictions through sequential refinement of residuals, captures the signal while linear models do not is consistent with the hypothesis that congressional trading interacts with market conditions in nonlinear ways. A trade by a senator on an oversight committee may matter more during periods of legislative activity, or the combination of high disclosure delay and coordinated buying may carry information that neither variable alone conveys.

A concern with any test of augmented models is that the improvement could reflect added flexibility rather than genuine information. Adding variables to a regularized model can improve in-sample fit through noise, and some of that noise may survive out of sample by chance. We address this through a placebo test: we replace the congressional features with the same number of randomly generated variables and re-estimate the augmented specification.

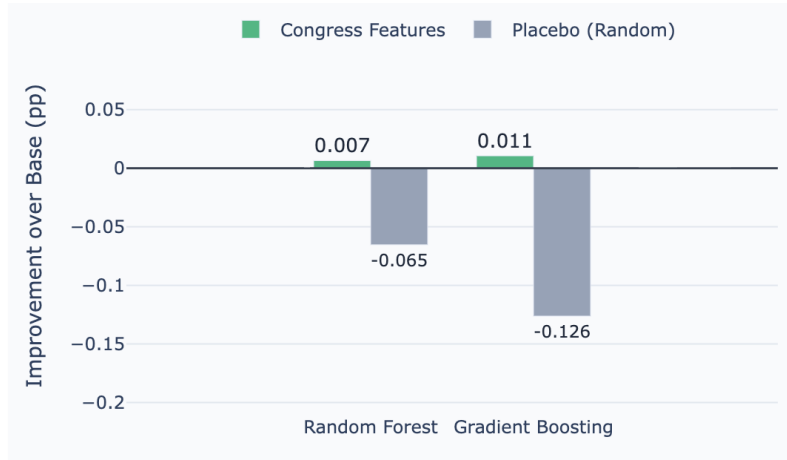


Figure 6: Improvement in R^2_{OOS} over the base model: congressional features versus placebo (random variables). The placebo improvement is negative for both models, confirming that the gains from congressional features are not an artifact of increased dimensionality.

Figure 6 shows the results of this test. The congressional features produce a positive improvement for both Random Forest and Gradient Boosting. The placebo features produce a substantially negative improvement in both cases. Random noise degrades prediction, as regularization alone cannot fully prevent uninformative variables from introducing distortion. The contrast is clear: the congressional features contain information

that the models can exploit, while adding the same number of irrelevant variables hurts performance.

4.3 What the Models Capture

The previous results establish that a signal contained in congressional data exists. We now examine what the models are learning and how the predictions relate to realized returns.

Figure 7 plots predicted versus actual cumulative abnormal returns for the augmented specification of both tree models. The predictions are tightly concentrated around zero, while actual returns exhibit much wider dispersion. The fitted line is nearly flat, with a correlation of approximately 0.12. However, this is not a failure of the model: in a setting where most return variation is driven by unforecastable news, a well-regularized model should produce conservative predictions. Attempting to match the full range of actual returns would require fitting noise, and the model correctly avoids this. The information content shows up not in the magnitude of predictions but in their ranking.

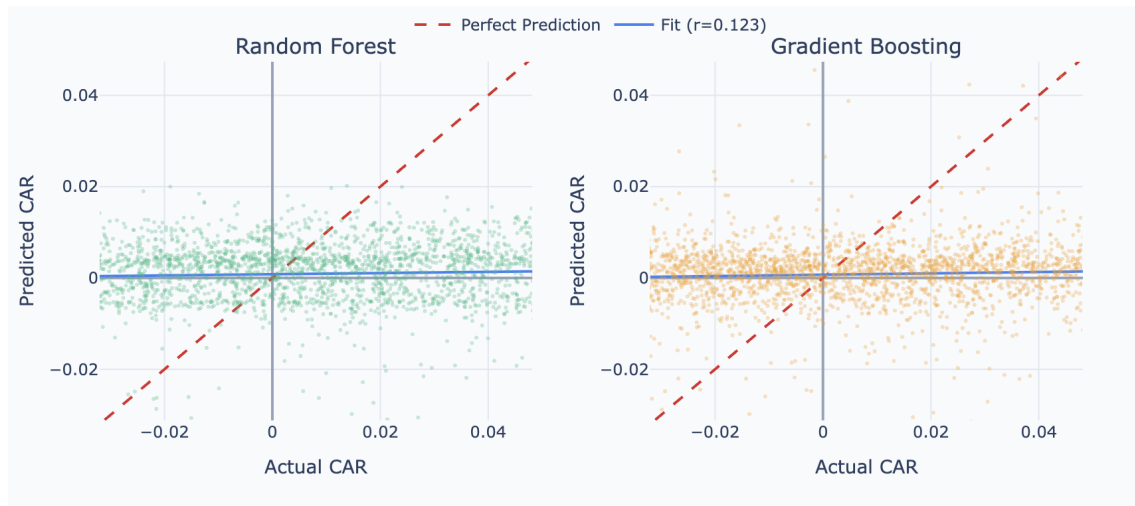


Figure 7: Predicted versus actual cumulative abnormal returns for Random Forest and Gradient Boosting (augmented specification). The dashed line indicates perfect prediction; the solid line is the linear fit.

This becomes apparent in Figure 8, which sorts observations into deciles by predicted return and then reports the average realized return within each decile. If the model captures no information, realized returns should be flat across deciles. If it captures signal, higher predicted deciles should correspond to higher realized returns. Both Random Forest and Gradient Boosting produce a monotonically increasing pattern. Stocks in the bottom decile of predicted returns realize average losses of approximately 2% over the subsequent 30-day window, while those in the top decile realize gains of similar magnitude. The spread between extreme deciles is roughly four percentage points, and the progression across intermediate deciles is smooth. This monotonicity is notable because the models

are not optimized for ranking—they minimize squared error—yet they learn to order stocks in a way that tracks realized performance.

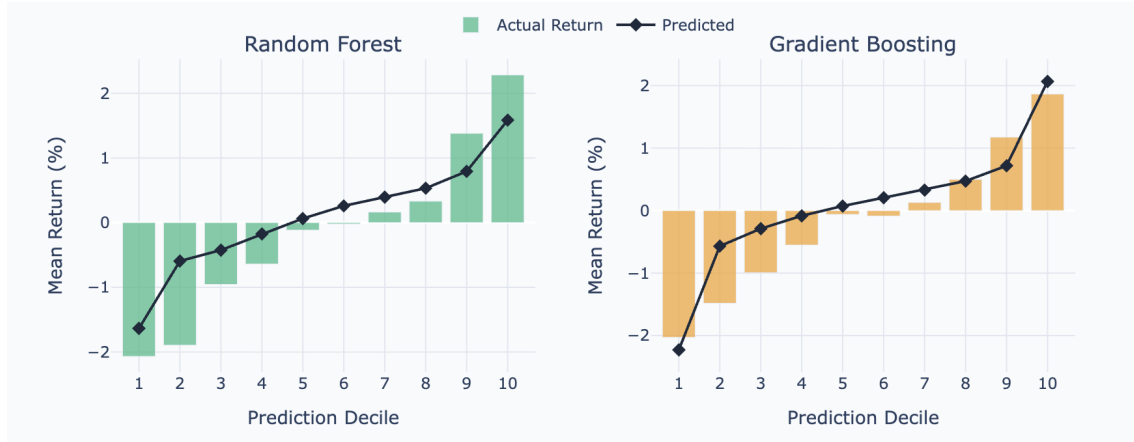


Figure 8: Average realized return by decile of predicted return. The black line shows the mean prediction within each decile; bars show the mean actual return. Both models exhibit monotonically increasing realized returns across prediction deciles.

Figure 9 shows the evolution of R_{OOS}^2 over time, computed on a rolling window of 2,000 observations. Predictive accuracy is not constant. All specifications exhibit periods of positive and negative R^2 , reflecting the changing informativeness of both market and congressional variables across different regimes. The augmented models tend to track above their base counterparts, though the gap varies. Predictability appears higher around 2020 and 2023, and lower during 2022, a period of rapid interest rate increases and elevated volatility that may have disrupted the usual relationship between political trading and returns. The instability of R_{OOS}^2 is itself consistent with the literature: Gu, Kelly, and Xiu (2020) document similar temporal variation in their cross-sectional return prediction exercise.

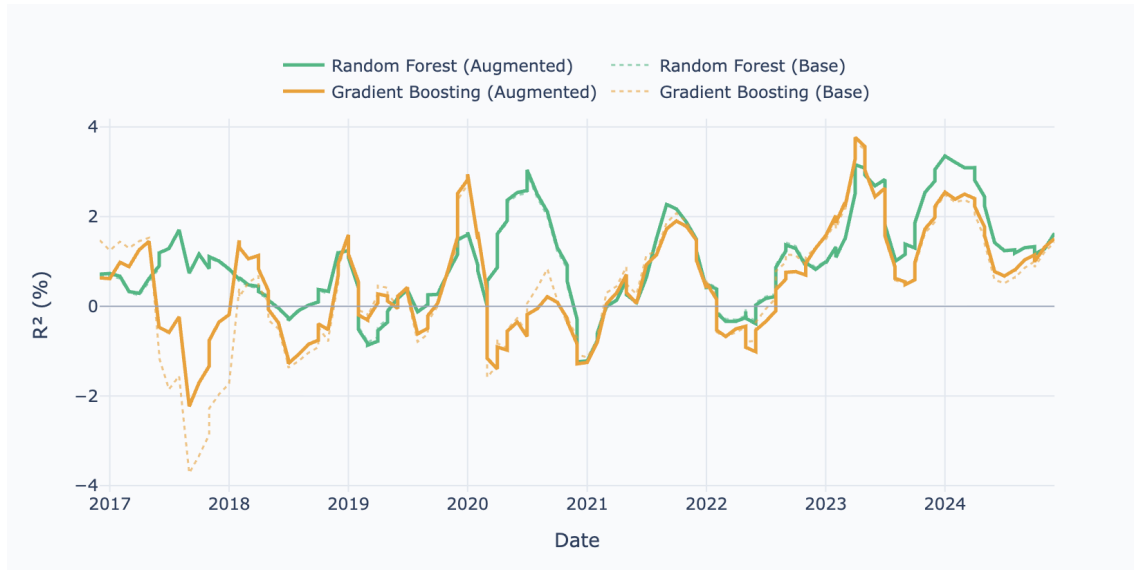


Figure 9: Rolling R^2_{OOS} over time (2,000-observation window). Solid lines show the augmented specification; dashed lines show the base.

Finally, we examine which congressional features the models rely on. Figure 10 reports permutation importance scores from the Random Forest augmented model, restricted to the congressional variables. The most informative feature is the average disclosure delay, that is the number of days between a trade's execution and its public filing. This is followed by the maximum disclosure delay and the average seniority of the politicians trading the stock. The information committee ratio, which measures the fraction of traders sitting on committees with oversight jurisdiction over the stock's sector, also ranks among the top predictors.

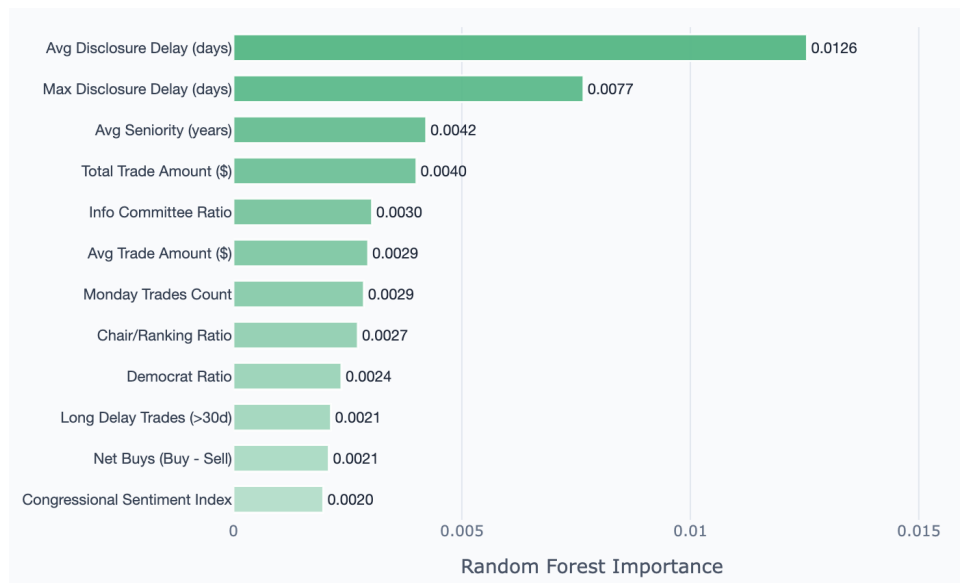


Figure 10: Permutation importance of congressional trading features in the Random Forest augmented model. Importance is measured as the decrease in R^2_{OOS} when each variable is randomly shuffled.

These rankings align with the theoretical priors outlined in the introduction. Disclosure delay is a measure of strategic timing: legislators who delay reporting may be trading on information they prefer not to reveal immediately. Seniority proxies for the depth of a politician’s network and the quality of their access to non-public information. Committee membership links a legislator’s institutional role to the specific firms they trade. The fact that the model assigns weight to these features rather than to simpler measures like the net direction of trading suggests that the predictive content is concentrated in the circumstances surrounding the trades, not merely in their direction. Whether a politician buys or sells matters less than who is trading, how quickly they disclose, and what institutional position they occupy.

5 Anomaly Detection

While the aggregate analysis in Section 4 evaluates whether congressional trades as a group contain predictive information for stock returns, this section adopts a complementary approach by identifying individual transactions that exhibit characteristics consistent with the potential use of material non-public information. This shift from aggregate prediction to individual trade detection addresses a fundamentally different question —not whether congressional trading predicts returns, but rather which specific trades display patterns suggestive of informed trading—.

The detection of potentially suspicious trading activity presents a methodological challenge that fundamentally differs from standard supervised learning problems. In supervised classification, one has access to labeled examples of both classes —in this context, trades known to be based on insider information and trades that are not—. The model learns to distinguish between classes by training on these labeled examples, and then applies the learned patterns to classify new observations. However, insider trading is clandestine: confirmed cases represent only a small fraction of actual instances, as most go undetected due to the difficulty of proving intent and the resource constraints of enforcement agencies.

Without reliable labels indicating which trades involve insider information, unsupervised learning methods become necessary. Unlike supervised approaches that learn from known examples, unsupervised anomaly detection algorithms identify observations that deviate significantly from the patterns exhibited by the majority of the data, without requiring prior knowledge of what constitutes an anomaly (Chandola, Banerjee, and Kumar 2009). The underlying assumption is that legitimate trading behavior, while heterogeneous, follows certain regularities that can be learned from data, whereas potentially suspicious trades will exhibit unusual combinations of characteristics that place them in low-density regions of the feature space.

Following the methodology proposed by Mazzarisi et al. (2024), who developed machine learning methods for insider trading detection, this analysis distinguishes two types of anomalies that capture different dimensions of suspicious trading behavior. Contextual anomalies are trades that would appear normal in isolation but are atypical given the specific context in which they occur. In this application, the relevant context is the historical trading pattern of the individual politician: a modest position in a technology stock might be unremarkable in general, but anomalous for a politician who has never previously traded in that sector. Collective anomalies are coordinated trading patterns where multiple politicians trade the same stock in the same direction within a short time window. While each individual trade might appear normal in isolation, the synchronized behavior suggests potential information sharing.

In addition to these two categories, this analysis also considers point anomalies: individual trades with extreme characteristics when compared to the universe of all trades, such as unusually large positions, trades in highly volatile stocks, or transactions with abnormally long disclosure delays. Across all three categories, only information available at the time of the trade is used.

The remainder of this section proceeds as follows. We first describe the methodology used to identify anomalous trades, employing multiple detection algorithms that capture different notions of what constitutes an outlier. A trade is classified as anomalous only if flagged by at least two of these methods, ensuring that our results are not driven by the idiosyncrasies of any single algorithm. The economic relevance of the detection approach is validated by comparing the cumulative abnormal returns of anomalous versus normal trades. Finally, the detected anomalies are characterized along several dimensions: the features that most distinguish them from normal trades, their distribution across parties, chambers, committees, and stock sectors, and their evolution over time.

It is necessary to emphasize from the outset that statistical anomaly detection does not constitute evidence of illegal conduct. The objective is not to adjudicate legality—which requires establishing intent and access to material non-public information—but rather to characterize which trades are statistically unusual and to evaluate whether these unusual trades exhibit superior returns consistent with the hypothesis of informed trading. If anomalous trades show no systematic outperformance relative to normal trades, this would suggest that our detection methods are capturing noise rather than information. Conversely, if anomalous trades generate significant abnormal returns, this provides indirect evidence that the methods are successfully identifying trades made with privileged information, though this cannot be determined from trading patterns alone.

5.1 Detection of Anomalous Trades

Following the classification introduced above, three types of anomalies are identified using four complementary algorithms, each capturing a different geometric notion of what constitutes an outlier. For the detection, 29 features were constructed capturing trade characteristics, market conditions at the time of the transaction, each politician’s historical trading pattern, and coordination among legislators. Each method independently flags trades as anomalous; a trade is then classified as anomalous in the final ensemble only if at least two methods agree. A contamination rate of 5% is assumed, meaning that roughly 5% of trades are expected to be anomalous—a standard threshold in the literature—. Table 2 summarizes the four detection methods, their target anomaly type, and the number of trades flagged as anomalous.

Table 2: Summary of Anomaly Detection Methods

Method	Anomaly Type	Contamination	Anomalies
Isolation Forest	Point	5%	4,004
One-Class SVM	Point	5%	7,338
LOF (per politician)	Contextual	5%	4,038
Rules + LOF	Collective	—	4,340
Ensemble (consensus ≥ 2)	All	—	4,144

Note: The rule-based component is deterministic; the LOF component uses 5% contamination.

To detect point anomalies, two algorithms are used to identify trades with extreme characteristics compared to the entire universe of congressional trading. Isolation Forest (Liu, Ting, and Zhou 2008) constructs an ensemble of random decision trees that recursively partition the feature space; anomalies, having extreme values, require fewer partitions to become isolated and thus receive higher anomaly scores. One-Class SVM (Schölkopf et al. 1999) takes a geometric approach, learning a decision boundary that encloses the normal region of the feature space and classifying observations outside this boundary as anomalies.

To detect contextual anomalies, Local Outlier Factor (LOF) (Breunig et al. 2000) is applied separately for each of the 139 politicians with at least 20 trades, identifying trades that are atypical relative to each politician’s own history. LOF measures the local density around each observation relative to its neighbors; observations in sparse regions receive high scores. By fitting a separate model per politician, each trade is compared against that individual’s historical pattern rather than against all congressional trades.

To detect collective anomalies, a hybrid approach combines rule-based criteria with statistical methods. The rule-based component flags a trade if: (i) at least three politicians traded the same ticker within a 7-day window, (ii) at least 70% traded in the same direction,

and (iii) the coordination is either bipartisan or involves members of the same committee. Additionally, LOF is applied to 7 collective features to detect unusual coordination patterns that may not satisfy the explicit rules. A trade is flagged as a collective anomaly if identified by either approach.

Each algorithm assigns an anomaly score to every trade —whether based on isolation depth, distance to a decision boundary, or local density— where higher values indicate greater deviation from normal patterns. These scores are then used to flag the top 5% as anomalous.

Rather than relying on any single algorithm, a trade is classified as anomalous if flagged by at least two of the four methods. This consensus approach reduces false positives by requiring agreement across methods with fundamentally different assumptions about what constitutes an outlier. Individually, Isolation Forest flags 4,004 trades (5.0%), contextual LOF flags 4,038 (5.1%), and the collective approach flags 4,340 (5.4%). One-Class SVM flags a higher proportion —7,338 trades (9.2%)— because its threshold is calibrated on a training subsample rather than the full dataset. When requiring consensus across at least two methods, the ensemble identifies 4,144 trades as anomalous, representing 5.2% of the 80,080 trades in the final sample.

Figure 11 provides visual validation of the detection approach. The left panel projects all 29 features onto two principal components using PCA and plots each trade, with anomalies identified by the ensemble highlighted in red. Anomalous trades cluster predominantly at the periphery of the point cloud rather than in the dense central region, confirming that the methods capture observations with unusual feature combinations. The right panel compares anomaly scores from Isolation Forest (horizontal axis) against those from contextual LOF (vertical axis). The correlation between these scores is moderate ($r = 0.34$), indicating that the two methods capture partially overlapping but distinct sets of anomalies. This is expected given their different objectives: Isolation Forest detects point anomalies —trades that are extreme relative to all congressional trading— while LOF detects contextual anomalies —trades that deviate from each politician’s personal historical pattern—. The partial overlap suggests the methods capture complementary dimensions of suspicious behavior.

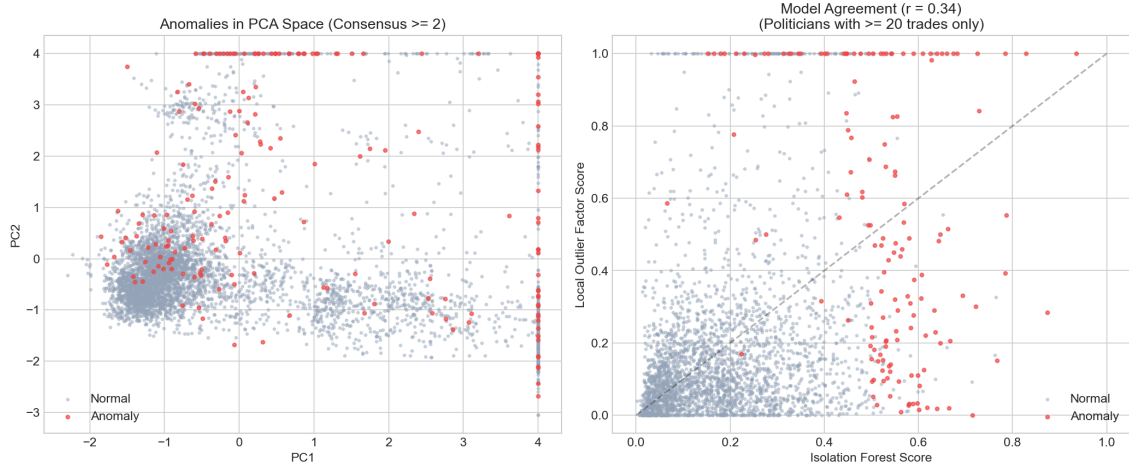


Figure 11: Anomaly detection. Left panel: trades projected onto the first two principal components, with anomalies (red) and normal trades (gray). Right panel: comparison of anomaly scores between Isolation Forest and Local Outlier Factor.

Figure 12 displays the distribution of normalized anomaly scores for three of the four detection methods, plus the ensemble score. All scores are scaled to the $[0, 1]$ interval using min-max normalization to enable comparison across methods. The dashed vertical line in each panel marks the 95th percentile, corresponding to the 5% contamination threshold used for flagging anomalies.

The top-left panel shows the Isolation Forest score distribution, which exhibits pronounced right skewness: the majority of trades cluster at low scores (below 0.4), indicating normal behavior, while a thin tail extends toward higher values representing potential anomalies. This shape is characteristic of well-calibrated anomaly detection, where most observations are correctly identified as normal. The top-right panel displays the contextual LOF distribution, computed only for the 71,610 trades from politicians with at least 20 transactions (excluding extreme values at the boundaries for visualization clarity). This distribution also exhibits right skewness, reflecting LOF's sensitivity to local density deviations within each politician's trading history.

The bottom-left panel presents the One-Class SVM score distribution, which appears more symmetric and centered around 0.25 compared to the other methods. This reflects the geometric nature of SVM, which measures distance from a learned decision boundary rather than isolation depth or local density. The bottom-right panel shows the ensemble score, computed as the arithmetic mean of the normalized scores. For trades from politicians with fewer than 20 transactions, the ensemble relies on the remaining three methods, as insufficient history precludes contextual LOF computation. This aggregation produces a smoother distribution that synthesizes information from all methods, with the 95th percentile threshold identifying the 4,144 trades (5.2%) classified as anomalous by the consensus rule.

Distribution of Anomaly Scores

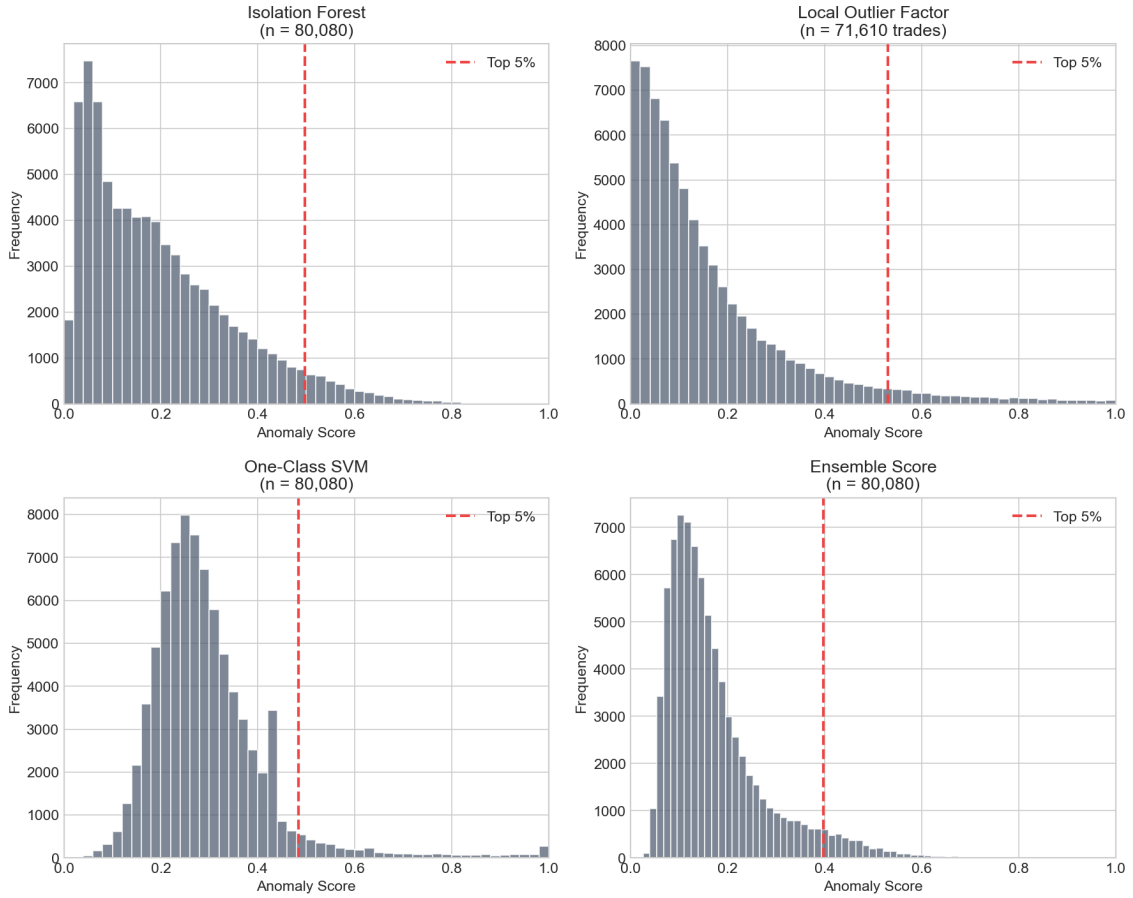


Figure 12: Distribution of normalized anomaly scores by detection method. The dashed vertical line indicates the 95th percentile threshold. Top left: Isolation Forest (point anomalies). Top right: Local Outlier Factor (contextual anomalies). Bottom left: One-Class SVM (point anomalies). Bottom right: Ensemble score (average of the normalized scores).

In summary, the detection approach identifies 4,144 trades (5.2%) as anomalous based on consensus across multiple methods. These trades exhibit unusual characteristics along different dimensions: extreme values relative to all congressional trading (point anomalies), deviations from each politician’s historical pattern (contextual anomalies), or coordination among multiple legislators (collective anomalies). The next section examines whether these statistically unusual trades also exhibit superior returns.

5.2 Validation: Abnormal Returns of Anomalous Trades

Once anomalous trades have been identified, a natural question arises: do these statistically unusual transactions exhibit superior performance consistent with the hypothesis of informed trading? To address this question, this section compares the returns of anomalous versus normal trades using Cumulative Abnormal Returns (CAR).

Abnormal returns are defined as the difference between the realized return of a stock and the return that would be expected given the stock's exposure to systematic risk factors. Following the standard approach in the asset pricing literature, expected returns are estimated using the Fama-French three-factor model (Fama and French 1993). This measure captures performance that cannot be attributed to exposure to known risk factors—precisely the type of return that would be expected if the trader possessed material non-public information—.

If the anomaly detection methods successfully identify trades made with superior information, anomalous trades should exhibit higher CAR than normal trades. Conversely, if the methods are merely capturing statistical noise, no systematic difference in returns should emerge. This comparison thus provides an economic validation of the detection approach.

Figure 13 presents the comparison of CAR between normal and anomalous trades across three panels. The left panel displays boxplots of the 30-day CAR distribution for each group. The median CAR for anomalous trades is higher than for normal trades, and this difference is statistically significant ($p\text{-value} < 0.001$).

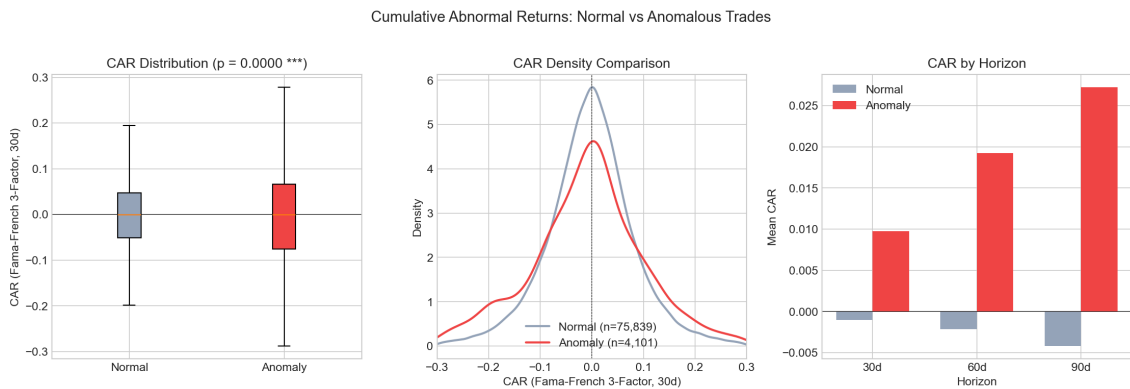


Figure 13: Cumulative Abnormal Returns (Fama-French 3-factor) for normal versus anomalous trades. Left panel: boxplot comparison. Center panel: kernel density estimates. Right panel: mean CAR by horizon (30, 60, 90 days).

The center panel shows kernel density estimates of the CAR distributions for both groups. The distribution for normal trades is centered approximately at zero, consistent with the efficient market hypothesis prediction that uninformed trades should not generate systematic abnormal returns. In contrast, the distribution for anomalous trades is shifted to the right, with greater probability mass in the positive return region.

The right panel examines how the CAR difference evolves across different time horizons (30, 60, and 90 days). Normal trades exhibit mean CAR close to zero or slightly negative at all horizons. Anomalous trades, however, show positive mean CAR that increases with the horizon: approximately 1% at 30 days, 2% at 60 days, and 2.7% at 90 days. This increasing pattern suggests that the information reflected in anomalous trades

takes time to be fully incorporated into prices. Table 3 reports the detailed comparison.

Table 3: CAR Comparison: Normal vs Anomalous Trades (Fama-French 3-Factor)

Horizon	Normal Mean	Anomaly Mean	Difference	t-statistic	p-value
30 days	−0.10%	+0.97%	+1.08 pp	5.94	< 0.001***
60 days	−0.21%	+1.92%	+2.14 pp	7.92	< 0.001***
90 days	−0.42%	+2.72%	+3.14 pp	8.54	< 0.001***

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3 shows that anomalous trades significantly outperform normal trades across all horizons. However, these results warrant careful interpretation due to a fundamental identification challenge inherent to anomaly detection methodologies.

The detection algorithms identify observations located in low-density regions of the feature space, where density is defined over the set of variables used for detection. Among these variables are market-based characteristics including realized volatility, systematic risk exposure (beta), and price momentum. A mechanical consequence of this approach is that trades involving stocks with extreme values on these dimensions are more likely to be classified as anomalous. This raises a concern: if anomalous trades are disproportionately concentrated in high-volatility or high-beta stocks, the observed outperformance may reflect compensation for systematic risk rather than the exploitation of private information.

A portfolio tilted toward high-beta or high-volatility stocks will, on average, generate higher returns than the market—not because of superior information, but because of greater risk bearing—. To address this concern, a robustness exercise re-estimates the anomaly detection models using exclusively political and behavioral features, such as disclosure delay, committee membership, coordination patterns among legislators, and trading behavior relative to each politician’s history. All market-based features are excluded. This ensures that trades flagged as anomalous are identified only on the basis of political characteristics that may proxy for informational advantage, rather than on stock-level attributes mechanically correlated with returns.

Table 4 reports the results. At the 30-day horizon, the CAR difference is economically small (0.32 percentage points) and not statistically significant ($p = 0.26$). However, at longer horizons, anomalies detected using only political features exhibit significantly higher returns: 1.04 percentage points at 60 days ($p = 0.011$) and 1.44 percentage points at 90 days ($p = 0.007$). This pattern—no effect at short horizons but significant outperformance at longer horizons—is consistent with the gradual incorporation of private information into prices (Kyle 1985).

Table 4: CAR Comparison: Detection with Political Features Only

Horizon	Normal Mean	Anomaly Mean	Difference	t-statistic	p-value
30 days	−0.05%	+0.28%	+0.32 pp	1.13	0.260
60 days	−0.09%	+0.95%	+1.04 pp	2.56	0.011**
90 days	−0.21%	+1.23%	+1.44 pp	2.72	0.007***

Note: Detection based exclusively on political features. Market-based features excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

These results suggest that the anomaly detection methods capture genuine informational content. While part of the baseline outperformance in Table 3 may reflect mechanical selection of high-volatility stocks, a significant effect emerges at longer horizons when detection relies exclusively on political characteristics. The absence of short-term outperformance combined with significant effects at 60 and 90 days is consistent with the hypothesis that some congressional trades are based on private information that is gradually incorporated into prices.

5.3 Characterization of Anomalous Trades

Having established that anomalous trades exhibit superior risk-adjusted returns, this section examines the characteristics that distinguish them from normal trades. The analysis reveals three consistent patterns: anomalous trades concentrate among politicians with greater access to market-sensitive information, cluster in sectors subject to regulatory uncertainty, and involve larger positions in more volatile and less liquid stocks.

Politicians with greater institutional access to information exhibit higher anomaly rates. Figure 14 shows that Republican members have a higher anomaly rate (6.5%) than Democrats (4.0%), senators higher than representatives (5.8% vs 5.1%), and committee leaders higher than regular members (5.5% vs 5.0%).

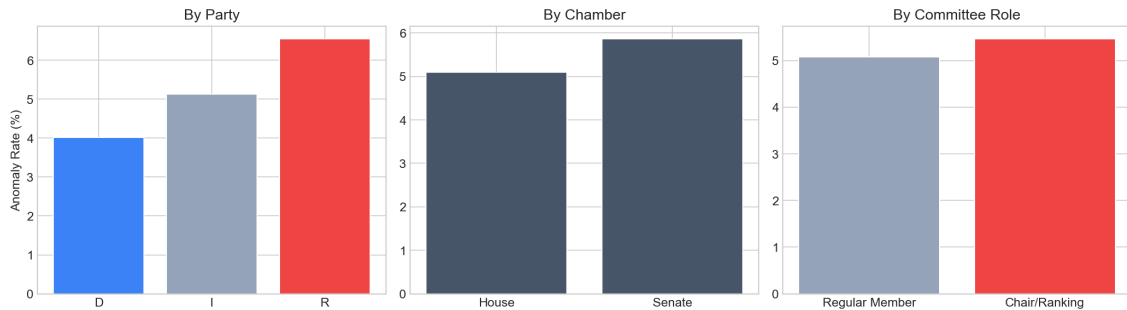


Figure 14: Anomaly rate by institutional characteristics. Left: party affiliation. Center: chamber. Right: committee role.

Figure 15 displays the individual anomaly rates for politicians with the highest pro-

portion of trades flagged as anomalous. The distribution is skewed toward Republican members, who account for 11 of the top 15 positions, compared to 4 Democrats. This asymmetry is consistent with the aggregate partisan differential documented in Figure 14 and suggests that the overall difference is not driven by a single outlier but rather by a systematic pattern across multiple legislators.

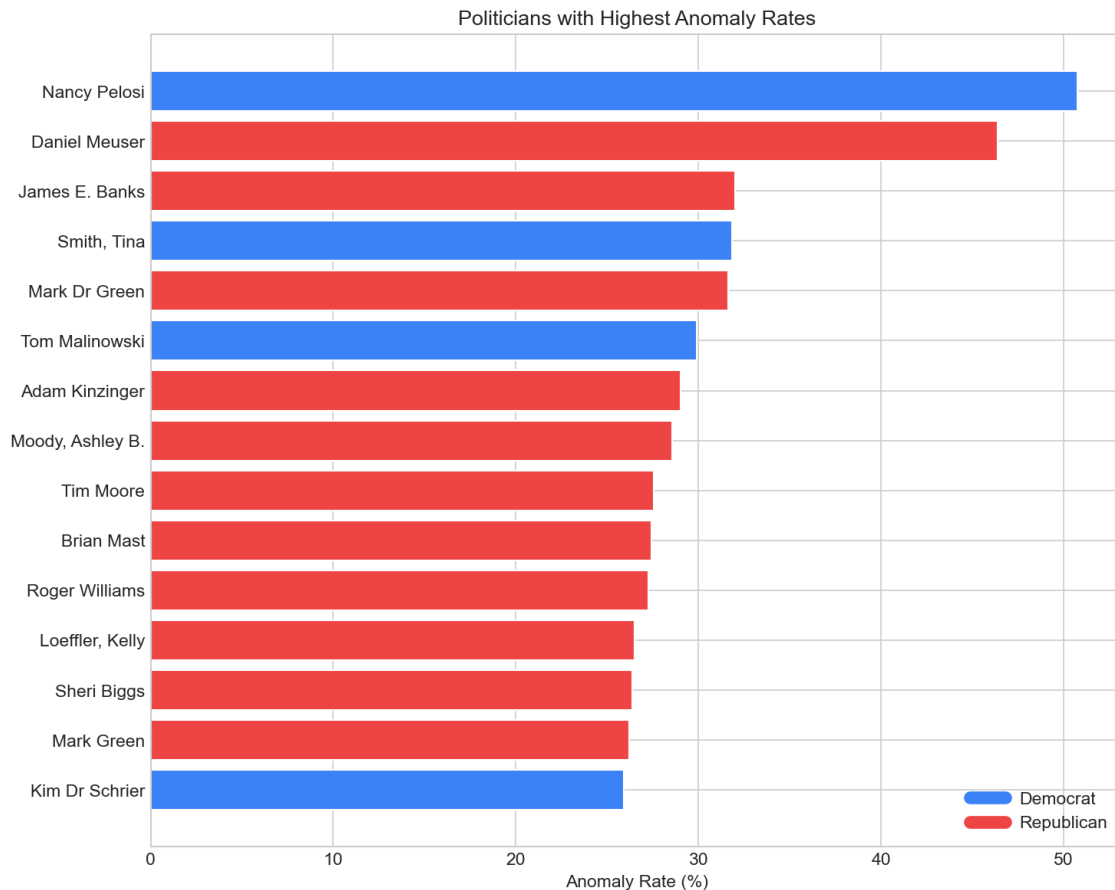


Figure 15: Politicians with highest anomaly rates.

The heterogeneity across committee assignments is even more striking. Figure 16 shows that the Commerce Committee exhibits the highest anomaly rate (15.9%), followed by Intelligence (14.8%), Judiciary (8.3%), and Finance (8.2%). These committees share a common characteristic: they oversee sectors with significant market capitalization and frequent regulatory activity, or have access to classified briefings on matters that can affect stock prices. In contrast, committees such as Budget (4.6%), Veterans Affairs (4.9%), and Energy (4.8%) show rates close to or below the sample average.

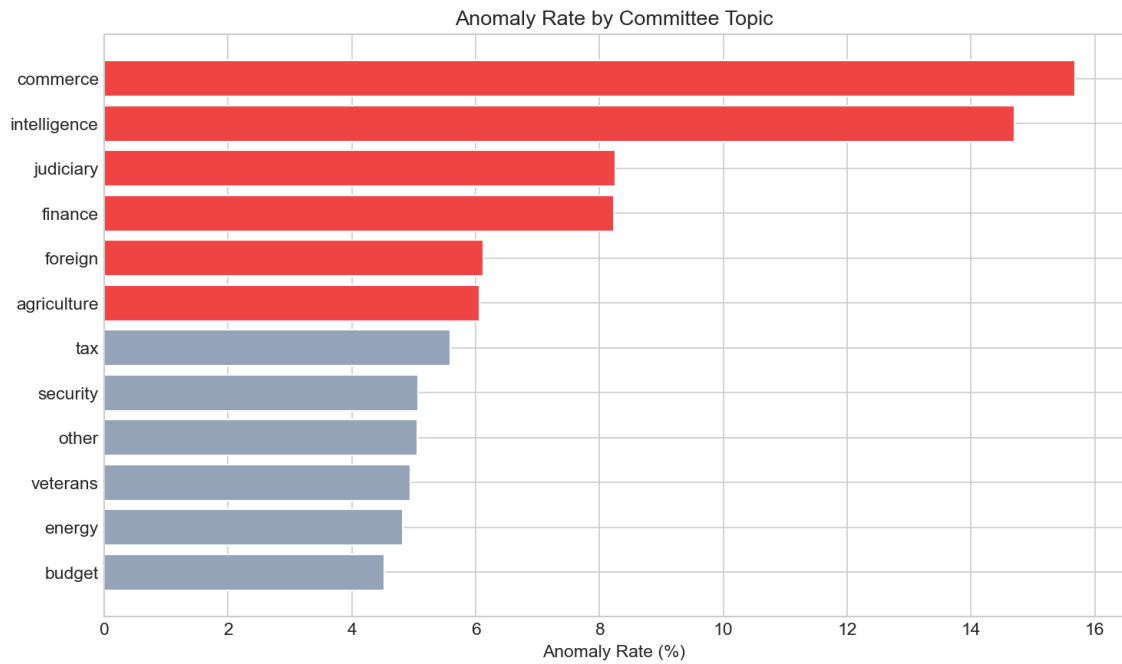


Figure 16: Anomaly rate by committee assignment.

A complementary pattern emerges in the sectoral distribution of anomalous trades. Figure 17 shows that Energy (7.0%), Technology (6.8%), and Communication Services (6.6%) exhibit the highest anomaly rates, while Consumer Defensive (2.8%) and Utilities (3.3%) show the lowest. This distribution is consistent with the committee findings: sectors characterized by higher volatility and frequent regulatory developments—precisely those overseen by the committees with elevated anomaly rates—show more unusual trading patterns.

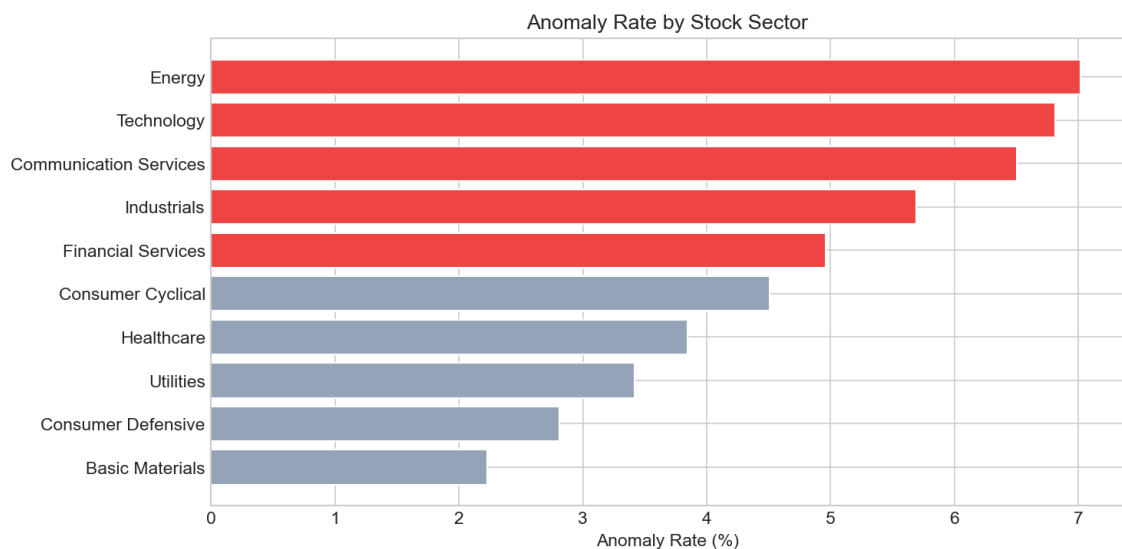


Figure 17: Anomaly rate by stock sector (GICS classification).

The feature profile of anomalous trades provides further support for the informed

trading hypothesis. Figure 18 presents Cohen’s d for each feature, revealing that anomalous trades are characterized by larger position sizes, higher stock volatility, lower liquidity, and elevated trading volume. This profile aligns with theoretical predictions: Kyle (1985) shows that informed traders optimally take larger positions to maximize profits, and that their advantage is more valuable in less liquid markets where prices have not yet incorporated the information.

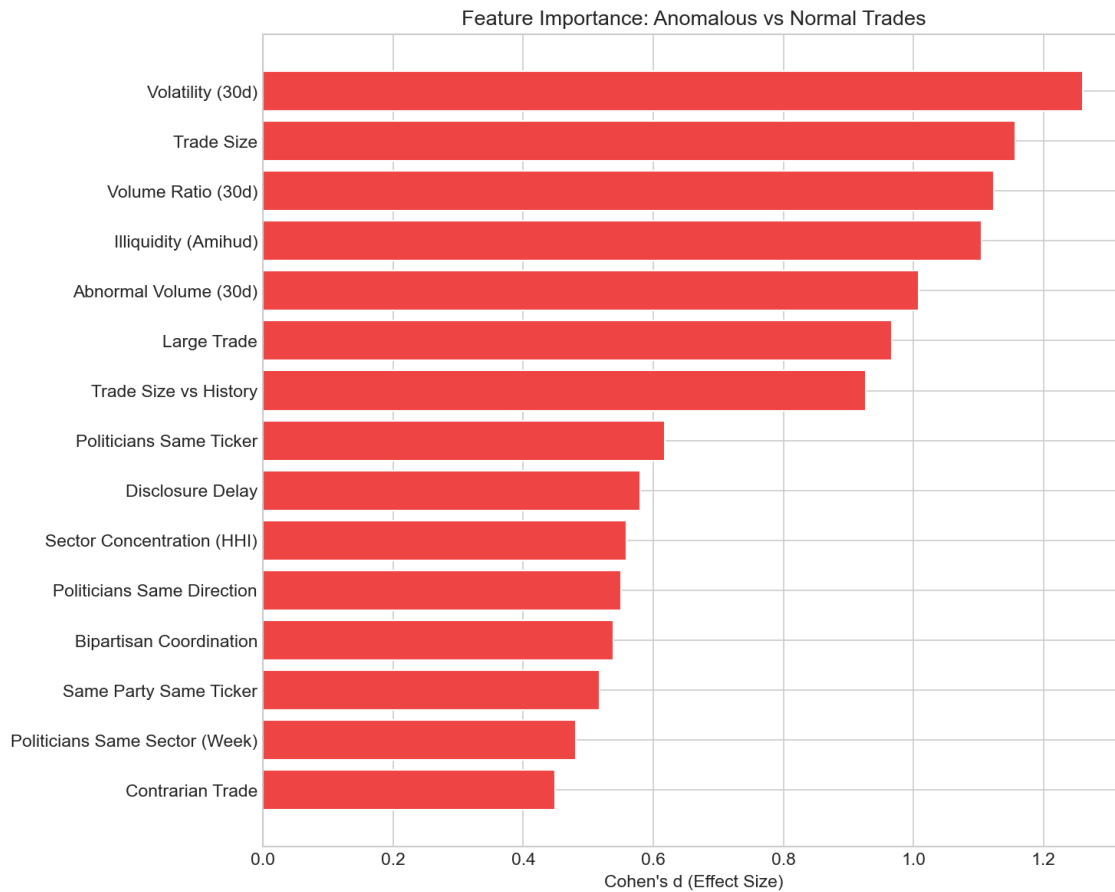


Figure 18: Feature importance measured by Cohen’s d (standardized mean difference between anomalous and normal trades).

Finally, the temporal evolution of anomaly rates provides external validation of the methodology. Figure 19 shows that the annual rate fluctuates between 3% and 6% for most of the sample period, with a notable spike to 11.5% in 2020. This spike coincides with the widely documented congressional trading scandal surrounding the COVID-19 pandemic. While this correspondence does not confirm that the detected anomalies represent illegal insider trading, the fact that the methods—which have no prior knowledge of these events— independently flag 2020 as an outlier year is consistent with the hypothesis that the methodology captures economically meaningful patterns.

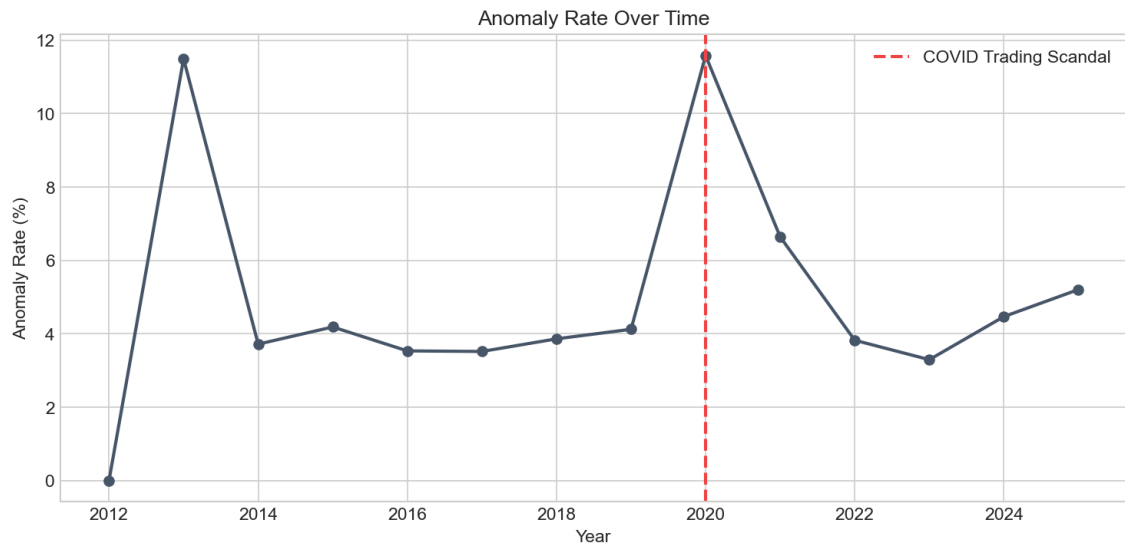


Figure 19: Anomaly rate by year.

In summary, anomalous trades are not randomly distributed. They concentrate among politicians with access to market-sensitive information, in sectors subject to regulatory uncertainty, and in stocks with characteristics favorable to informed trading. These mutually reinforcing patterns strengthen the interpretation that the detected anomalies reflect, at least in part, trading informed by non-public information.

6 Conclusion

We developed a two-stage empirical framework to assess whether U.S. congressional stock trades contain private information. The first stage used supervised learning to test whether adding congressional trading variables to standard market predictors improves out-of-sample return forecasts. The second stage used unsupervised anomaly detection to identify individual trades with unusual characteristics and then evaluated whether those trades earned abnormal returns. The two stages address different questions but produce mutually reinforcing evidence: the supervised analysis establishes that congressional trading carries some aggregate predictive content, while the unsupervised analysis shows that this content is concentrated among trades with specific institutional features —high disclosure delays, traders on oversight committees, and coordinated positions across legislators—. The fact that anomalous trades generate significant abnormal returns at longer horizons, even when detection relies exclusively on political rather than market features, suggests that the signal is not an artifact of mechanical risk exposure but reflects genuine informational asymmetry.

From a methodological standpoint, the finding that only nonlinear models capture the congressional signal has practical implications for future work in this area. Linear specifications, including regularized ones, failed to detect any contribution from the political

variables —not because the variables are irrelevant, but because their relevance operates through interactions and thresholds that linear functional forms cannot represent—. This underscores the value of combining economic theory with flexible estimation: theory guided the construction of 70 features that would have been difficult to discover algorithmically, while tree-based methods captured relationships among those features that would have been impossible to specify manually. Several limitations should be acknowledged. The STOCK Act’s reporting in amount ranges rather than exact values introduces measurement error, the sample is conditioned on stocks that politicians chose to trade, and statistical anomalies are not proof of illegal conduct. Nonetheless, the consistency of the results across both stages of the analysis, the robustness to placebo tests, and the alignment between detected anomalies and known episodes of suspicious trading suggest that the framework provides a useful tool for monitoring legislative trading activity and informing the ongoing debate about the adequacy of current disclosure requirements.

Bibliography

- Breunig, Markus M et al. (2000). “LOF: identifying density-based local outliers”. In: *Proceedings of the 2000 ACM SIGMOD international conference on Management of data*, pp. 93–104.
- Campbell, John Y and Samuel B Thompson (2008). “Predicting excess stock returns out of sample: Can anything beat the historical average?” In: *The Review of Financial Studies* 21.4, pp. 1509–1531.
- Chandola, Varun, Arindam Banerjee, and Vipin Kumar (2009). “Anomaly detection: A survey”. In: *ACM computing surveys (CSUR)* 41.3, pp. 1–58.
- Clark, Todd E and Kenneth D West (2007). “Approximately normal tests for equal predictive accuracy in nested models”. In: *Journal of Econometrics* 138.1, pp. 291–311.
- Fama, Eugene F and Kenneth R French (1993). “Common risk factors in the returns on stocks and bonds”. In: *Journal of financial economics* 33.1, pp. 3–56.
- Ferraz, Claudio and Frederico Finan (2008). “Exposing corrupt politicians: the effects of Brazil’s publicly released audits on electoral outcomes”. In: *The Quarterly journal of economics* 123.2, pp. 703–745.
- Grossman, Sanford J and Joseph E Stiglitz (1980). “On the Impossibility of Informationally Efficient Markets”. In: *The American Economic Review* 70.3, pp. 393–408.
- Gu, Shihao, Bryan T Kelly, and Dacheng Xiu (2020). “Empirical asset pricing via machine learning”. In: *The Review of Financial Studies* 33.5, pp. 2223–2273.
- Karadas, Serkan (2019). “Trading on private information: Evidence from members of Congress”. In: *Financial Review* 54.1, pp. 85–131.

- Karadas, Serkan, Minh Tam Tammy Schlosky, and Joshua Hall (2021). “Did politicians use non-public macroeconomic information in their stock trades? Evidence from the STOCK Act of 2012”. In: *Journal of Risk and Financial Management* 14.6, p. 256.
- Kyle, Albert S (1985). “Continuous auctions and insider trading”. In: *Econometrica: Journal of the Econometric Society*, pp. 1315–1335.
- Liu, Fei Tony, Kai Ming Ting, and Zhi-Hua Zhou (2008). “Isolation forest”. In: *2008 eighth ieee international conference on data mining*. IEEE, pp. 413–422.
- Mazzarisi, Piero et al. (2024). “A machine learning approach to support decision in insider trading detection”. In: *EPJ Data Science* 13.1, p. 66.
- Moe, Ye Brandon and Yao Ma (2024). “Congressional Stock Trades and Economic Policy Uncertainty”. In: *FEB-RN Research Paper* 13.
- Mullainathan, Sendhil and Jann Spiess (2017). “Machine learning: an applied econometric approach”. In: *Journal of Economic Perspectives* 31.2, pp. 87–106.
- Schölkopf, Bernhard et al. (1999). “Support vector method for novelty detection”. In: *Advances in neural information processing systems* 12.
- Snyder Jr, James M and David Strömberg (2010). “Press coverage and political accountability”. In: *Journal of political Economy* 118.2, pp. 355–408.
- Welch, Ivo and Amit Goyal (2008). “A comprehensive look at the empirical performance of equity premium prediction”. In: *The Review of Financial Studies* 21.4, pp. 1455–1508.
- Ziobrowski, Alan J, James W Boyd, et al. (2011). “Abnormal returns from the common stock investments of members of the US House of Representatives”. In: *Business and Politics* 13.1, pp. 1–22.
- Ziobrowski, Alan J, Ping Cheng, et al. (2004). “Abnormal returns from the common stock investments of the US Senate”. In: *Journal of financial and quantitative analysis* 39.4, pp. 661–676.